

Data Plane — WireGuard core, pluggable transports, obfuscation, routing

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Master technical specification — document 01 of the HelixVPN set. Scope: the **Rust data plane** shared byte-for-byte by client, connector, and gateway edge. This is a SPEC (describe the implementation; do not build the product). Source evidence cited inline by id, e.g. [04_ARCH §3], [04_P0 §4.2], [11_MST], [SYNTHESIS].

0. Position in the system & what this document owns

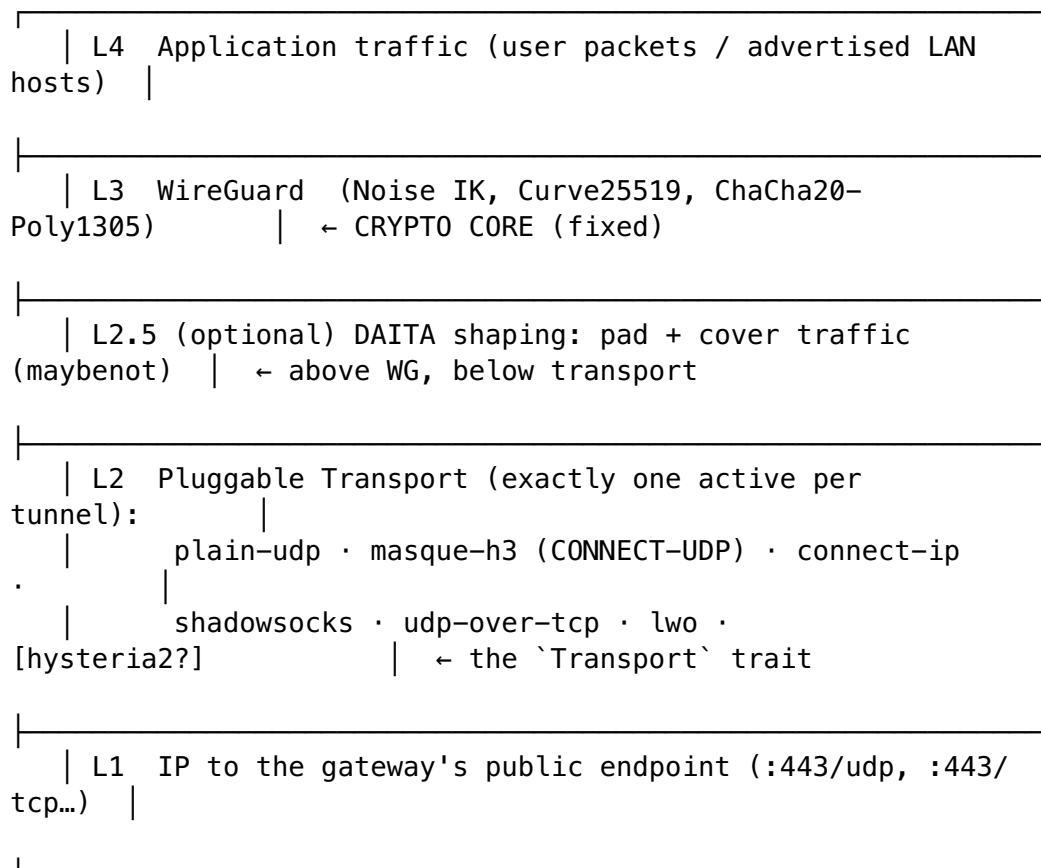
HelixVPN is a self-hostable overlay network with a privacy-VPN front end: three roles — **Connector** (network-side outbound-only agent advertising CIDRs) $\rightleftharpoons$ **Gateway** (public VPS: control + data plane) $\rightleftharpoons$ **Client** (Mullvad-style end-user app) [04_ARCH §1.1, 05_YBO, SYNTHESIS §1]. This document specifies **only the data plane** — the code that moves already-encrypted bytes — which is the same Rust workspace on all three roles [04_ARCH §3.2, §5.5]. It does **not** specify the Go control plane (doc 02), the WatchNetworkMap protobuf (doc 03), client UI (doc 05), or platform tunnel shims beyond the TUN handoff contract (doc 06).

The single most important architectural constraint that governs every line below [04_ARCH §0, §2.2]: *WireGuard is the cryptographic core everywhere; obfuscation/transport is a swappable layer **under** WireGuard, never a fork of WG crypto.* Mullvad's "QUIC mode" is not a separate protocol — it is WireGuard tunnelled over MASQUE/HTTP-3 [04_ARCH §0, research synthesis D1]. HelixVPN keeps WG as the L3 crypto and treats QUIC/Shadowsocks/ UDP-over-TCP/LWO as interchangeable L2 carriers of opaque WG datagrams.

0.1 Non-negotiable data-plane invariants

#	Invariant	Source
I1	The transport layer never sees plaintext ; it carries only already-encrypted WG datagrams.	[04_P0 §4.2]
I2	The transport carries unreliable datagrams , never an ordered byte stream — preserving WG's own loss semantics, avoiding head-of-line blocking.	[04_P0 §4.2 design note]
I3	Control plane is never in the packet path ; if control is down, existing tunnels keep forwarding (fail-static).	[04_ARCH §2.1]
I4	One transport crate, three consumers (client, connector, edge): the code that obfuscates on the client is the same code that de-obfuscates on the edge.	[04_ARCH §3.2, §5.5]
I5	No-logging by construction in the data plane: only aggregate counters, never per-connection/per-packet durable state.	[04_ARCH §2.7, §7]
I6	Default-deny : no peer reaches anything without an explicit compiled policy rule expressed as AllowedIPs + an edge verdict map.	[04_ARCH §3.4, §7]

1. Layering model



WireGuard owns confidentiality, integrity, and roaming. The transport layer **only changes how encrypted WG datagrams look on the wire** so DPI cannot fingerprint or block them [04_ARCH §3.1]. DAITA is orthogonal to *which* transport runs (works with plain-udp or masque alike) [04_P2 §2.2] and is specified at §9.

2. Crate layout (the data-plane workspace)

The Phase-0 interfaces are designed to **survive into Phase 1** [04_P0 §4]. The workspace is a decoupled, reusable Rust component per constitution §11.4.28/.74 (its own vasic-digital repo, snake_case, flat submodule) [SYNTHESIS §9].

```

helix-core/                                # Cargo workspace; repo:
vasic-digital/helix_core
├─ Cargo.toml                             # [workspace]
├─ crates/
│   └─ helix-transport/                   # the pluggable L2 carrier –
shared client+edge+connector
│   │   └─ src/lib.rs                     # Transport trait +
TransportConfig + dial() + registry
│   │   └─ src/plain_udp.rs               # §3.2
│   │   └─ src/masque.rs                  # §3.3 (quinn + h3 +
CONNECT-UDP/HTTP-Datagram)
│   │   └─ src/connect_ip.rs              # §3.4 (RFC 9484, advanced
– no inner WG)
│   │   └─ src/shadowsocks.rs             # §3.5
│   │   └─ src/udp_over_tcp.rs            # §3.6
│   │   └─ src/lwo.rs                     # §3.7
│   │   └─ src/hysteria2.rs               # §3.8 (option, decision
D1)
│   │   └─ src/error.rs                   # TransportError taxonomy
│   └─ helix-wg/                          # §4 boringtun wrapper:
handshake, encrypt/decrypt, timers
│   └─ helix-tun/                          # OS TUN abstraction (Linux
native; shims inject the fd)
│   └─ helix-daita/                        # §9 maybenot shaping stage
│   └─ helix-route/                       # §6–§8 overlay addressing,
4via6, ACL→verdict-map compiler
│   └─ helix-core/                         # §5 orchestrator: ties tun ⇌
wg ⇌ daita ⇌ transport; status
│   └─ helix-ffi/                         # flutter_rust_bridge surface
(doc 05)
└─ bin/
    └─ helix-client.rs                    # Linux CLI client
    └─ helix-connector.rs                 # Linux CLI connector
(advertise/route mode)
    └─ helix-edge.rs                      # gateway data-plane edge
(MASQUE termination) – see D5

```

The same helix-edge binary on the gateway is the third consumer of helix-transport [04_ARCH §11, SYNTHESIS §6]. Decision D5 (Rust vs Go edge) is settled by the Phase-0 G4 benchmark (§11); this spec assumes the **Rust** edge for the single-implementation guarantee and notes the Go fallback at §11.

3. The Transport trait — the single most important interface

Everything obfuscation-related hides behind one trait. `plain_udp`, `masque`, ... implement it; the client, connector, and edge all consume it. **One trait, three consumers, N transports** [04_P0 §4.2].

3.1 Trait, config, error, and the `dial()` ladder

```
// helix-transport/src/lib.rs
use async_trait::async_trait;
use bytes::Bytes;
use std::net::SocketAddr;

/// A bidirectional carrier for ALREADY-ENCRYPTED WireGuard
/// datagrams.
/// The transport NEVER sees plaintext (I1); it carries unreliable
/// datagrams,
/// never an ordered stream (I2). Implementations are cancel-safe.
#[async_trait]
pub trait Transport: Send + Sync {
    /// Send exactly one WG datagram toward the peer endpoint.
    async fn send(&self, datagram: Bytes) -> Result<(),
        TransportError>;

    /// Receive the next WG datagram from the peer (cancel-safe;
    /// usable in `select!`).
    async fn recv(&self) -> Result<Bytes, TransportError>;

    /// Stable label for logs/metrics: "plain-udp", "masque-h3",
    /// "shadowsocks", ...
    fn kind(&self) -> &'static str;

    /// MTU the upper layer (WG) may use over this transport,
    /// AFTER per-transport overhead (§10).
    fn effective_mtu(&self) -> u16;

    /// Snapshot of liveness (RTT EWMA, last-recv age) the
    /// orchestrator uses for ladder decisions.
    fn health(&self) -> TransportHealth;

    /// Graceful close (flush QUIC, FIN the TCP carrier).
    /// Idempotent.
    async fn close(&self) -> Result<(), TransportError>;
}
```

```

#[derive(Clone, Debug)]
pub struct TransportHealth {
    pub rtt_ewma_ms: Option<u32>,
    pub last_rcv_age_ms: u64,
    pub send_errors: u32,
}

/// One variant per L2 carrier. The escalation ladder constructs
the next
/// variant on repeated handshake failure (§5.3). All variants
resolve their
/// concrete endpoint + secrets from the NetworkMap pushed by the
coordinator (doc 03).
#[derive(Clone, Debug)]
pub enum TransportConfig {
    PlainUdp { peer: SocketAddr, bind: SocketAddr },
    MasqueH3 { url: String, sni: String, bind: SocketAddr,
               congestion: Congestion },
    ConnectIp { url: String, sni: String, bind:
                SocketAddr }, // RFC 9484, advanced (§3.4)
    Shadowsocks { peer: SocketAddr, method: SsMethod, psk:
                  SecretBytes },
    UdpOverTcp { peer: SocketAddr, tls_sni: Option<String> },
    Lwo { peer: SocketAddr, bind: SocketAddr,
          session_key: SecretBytes },
    #[cfg(feature = "hysteria2")]
    Hysteria2 { url: String, sni: String, salamander_pw:
                 SecretBytes }, // decision D1 (§3.8)
}

#[derive(Clone, Copy, Debug)] pub enum Congestion { Cubic, Bbr }
#[derive(Clone, Copy, Debug)] pub enum SsMethod {
    Chacha20Poly1305, Aes256Gcm }

/// Build a live transport from config. Returns within a bounded
dial timeout;
/// on `DialTimeout`/`HandshakeFailed` the orchestrator escalates
to the next rung (§5.3).
pub async fn dial(cfg: TransportConfig) -> Result<Box<dyn
    Transport>, TransportError> { /* ... */ }

// helix-transport/src/error.rs
#[derive(thiserror::Error, Debug)]
pub enum TransportError {
    #[error("dial timed out")] DialTimeout,
    #[error("handshake failed: {0}")]
        HandshakeFailed(String), // → triggers ladder escalation
    #[error("peer endpoint blocked")]
        EndpointBlocked, // DPI/firewall verdict
    #[error("transport closed")] Closed,
    #[error("oversize datagram {0} > mtu")] Oversize(usize),
    #[error("io: {0}")] Io(#[from]
        std::io::Error),
    #[error("quic: {0}")] Quic(String),
}

```

Design rationale (I2): because WG datagrams are independent and loss-tolerant, the transport carries unreliable datagrams. For MASQUE this means **QUIC DATAGRAM frames (RFC 9221)** carried as **HTTP Datagrams (RFC 9297)**, *not* a QUIC stream — preserving WG's loss semantics and avoiding head-of-line blocking [04_P0 §4.2, §5.1]. This is the half of S3 that earns Phase 0.

3.2 plain-udp — the baseline (Phase 0 / always available)

The default, lowest-latency, lowest-CPU path on unrestricted networks [04_ARCH §3.2]. It also sets the throughput baseline every other transport is measured against ($\geq 80\%$ of bare link is the G1 gate) [04_P0 §0, §8].

```
// helix-transport/src/plain_udp.rs
pub struct PlainUdp { sock: tokio::net::UdpSocket, peer:
    SocketAddr, health: HealthCell }

#[async_trait]
impl Transport for PlainUdp {
    async fn send(&self, dg: Bytes) -> Result<(), TransportError>
    {
        self.sock.send_to(&dg, self.peer).await?; Ok(())
    }
    async fn recv(&self) -> Result<Bytes, TransportError> {
        let mut buf = vec![0u8; 1500];
        let n = self.sock.recv(&mut buf).await?;
        self.health.mark_recv();
        buf.truncate(n); Ok(Bytes::from(buf))
    }
    fn kind(&self) -> &'static str { "plain-udp" }
    fn effective_mtu(&self) -> u16 { 1420 } // standard WG-
        over-IPv4 (§10)
    fn health(&self) -> TransportHealth { self.health.snapshot() }
    async fn close(&self) -> Result<(), TransportError> { Ok(()) }
}
```

3.3 masque-h3 — CONNECT-UDP over HTTP/3 (the headline obfuscation)

The "QUIC mode" — Mullvad's actual mechanism, WG-over-MASQUE [04_ARCH §0, §3.3]. The client wraps each WG datagram in an HTTP/3 **CONNECT-UDP** flow (RFC 9298) to the gateway's :443/udp HTTP/3 listener; to a passive observer the flow is indistinguishable from a browser doing HTTP/3 [04_ARCH §3.3, 04_P0 §5].

client WG datagram (Bytes)

↳ HTTP Datagram (RFC 9297, context-id = 0 for the proxied UDP payload)

↳ QUIC DATAGRAM frame (RFC 9221, unreliable – matches WG semantics, I2)

↳ QUIC / HTTP-3 connection to https://gateway:443 (looks like web)

↳ EDGE: extract HTTP Datagram → WG datagram → kernel WG fast path

RFC stack: **RFC 9298** (CONNECT-UDP / Proxying UDP in HTTP), **RFC 9297** (HTTP Datagrams & Capsule Protocol), **RFC 9221** (unreliable QUIC datagrams) [04_P0 §5.1, research-masque]. The CONNECT-UDP request establishes the proxied UDP "flow" once at `dial()`; thereafter WG datagrams ride as HTTP Datagrams with **no per-packet HTTP round trip** [04_P0 §5.1].

```
// helix-transport/src/masque.rs (the parts that matter)
pub struct MasqueTransport {
    conn: quinn::Connection,           // established QUIC/H3
        connection to gw:443
    flow_ctx: u64,                     // CONNECT-UDP context-id
        (0) established at dial()
    health: HealthCell,
}

#[async_trait]
impl Transport for MasqueTransport {
    async fn send(&self, wg: Bytes) -> Result<(), TransportError>
    {
        let http_dg = encode_http_datagram(self.flow_ctx,
            &wg); // RFC 9297 framing
        self.conn.send_datagram(http_dg).map_err(|e|
            TransportError::Quic(e.to_string()))?;

        Ok(()) // RFC 9221 QUIC datagram
    }

    async fn recv(&self) -> Result<Bytes, TransportError> {
        let dg = self.conn.read_datagram().await.map_err(|e|
            TransportError::Quic(e.to_string()))?;
        self.health.mark_recv();

        Ok(decode_http_datagram(dg)?) // strip framing -> WG bytes
    }

    fn kind(&self) -> &'static str { "masque-h3" }
    fn effective_mtu(&self) -> u16 { 1280 } // QUIC overhead
        eats headroom - measure & tune (§10)
    fn health(&self) -> TransportHealth { self.health.snapshot() }
    async fn close(&self) -> Result<(), TransportError> {
        self.conn.close(0u32.into(), b"bye"); Ok(()) }
}
```

Building blocks & maturity caveat [04_P0 §5.2]: quinn (mature QUIC, exposes `send_datagram/read_datagram`); h3 (HTTP/3 on quinn, less battle-tested than Go's stack); **MASQUE CONNECT-UDP handling is thin-to-absent as a turnkey Rust crate** — expect to implement CONNECT-UDP request + HTTP-Datagram framing by hand on top of h3 + quinn datagrams. This relative immaturity vs Go's `masque-go` is itself the input to decision D5 (edge language, §11).

Masquerade [04_ARCH §3.3, 04_P0 §5.4]: the edge :443 listener serves a believable decoy site to anything that is *not* a valid CONNECT-UDP flow (probes/scanners) — native edge behavior replacing the original doc's Nginx-camouflage. Verify the flow classifies as HTTP/3 with no WG signature via `tshark` (the G2 wire-fingerprint check) [04_P0 §8].

Loss resilience is the reason mobile got this feature: under netem loss 5%, masque/QUIC must sustain higher goodput than the UDP-over-TCP strawman [04_P0 §5.3]. Congestion control is selectable (Cubic default; Bbr for lossy mobile — the Hysteria "Brutal" lineage tuning knob is a useful reference for quinn window ratios) [04_ARCH §14].

3.4 connect-ip — IP-over-HTTP/3 (RFC 9484, advanced)

An alternative datapath that proxies **IP packets** (not inner WG) directly over HTTP/3 via **CONNECT-IP** [04_ARCH §3.2, research-masque]. Use when a native IP-over-H3 path is desired without the inner WG layer (e.g. the gateway acts as a true IP router for the flow). **Spec stance:** ship behind a feature flag in Phase 2+; the default and recommended path remains inner-WG-over-masque-h3 (§3.3) so the crypto core (WG) and the no-logging/ACL model are unchanged. CONNECT-IP loses I1's "transport never sees plaintext" property at the gateway (the gateway terminates IP), so it is **not** offered to privacy clients by default — only for explicit site-to-site routing where the gateway is already a trusted router. Recommendation: implement after masque-h3 is proven; do not block the MVP on it.

3.5 shadowsocks — WG-in-Shadowsocks (China-grade DPI)

WG datagrams wrapped in a Shadowsocks AEAD stream — the canonical "looks like random/TLS-ish TCP" evasion where even QUIC is throttled [04_ARCH §3.2, 04_P2 §1.1].

```
// helix-transport/src/shadowsocks.rs (sketch)
pub struct ShadowsocksTransport {
    stream: shadowsocks_crypto::AeadStream, // chacha20-
        poly1305 / aes-256-gcm
    // session keys derived from a pre-shared transport password,
    // SEPARATE from WG keys
}
// send: frame WG datagram with 2-byte length prefix → AEAD
//        encrypt → TCP
// recv: AEAD decrypt → deframe → WG datagram
```

Reuse shadowsocks-rust crypto primitives rather than re-implementing AEAD framing [04_P2 §1.1]. Because it rides TCP it carries the head-of-line-blocking caveat; the ladder prefers it only when UDP/QUIC are blocked. `effective_mtu()` reports a conservative value to leave room for the 2-byte length prefix + AEAD tag (§10).

3.6 udp-over-tcp — last resort when all UDP is blocked

WG datagrams length-prefixed over a single TCP connection (matches Mullvad's `udp2tcp`) [04_ARCH §3.2, 04_P2 §1.2]. Accept the head-of-line-blocking penalty; it exists purely to keep a tunnel *possible* on the most hostile networks. Optional TLS wrap (`tls_sni`) to look like HTTPS. Lowest rung of the ladder.

3.7 lwo — lightweight WG obfuscation (cheap evasion)

Per-session keyed obfuscation of the WG header bytes that DPI signatures key on (message-type / reserved fields) plus randomized padding — near-zero-cost evasion of *naive* WG fingerprinting without QUIC overhead [04_ARCH §3.2, 04_P2 §1.3]. Phase 1 ships a basic XOR/padding scheme; Phase 2 hardens it to a proper per-session keyed scheme. Mullvad's "LWO" is the design reference. It is the **first** escalation rung above plain-udp because it is the cheapest non-default option.

3.8 hysteria2 — decision D1 (option, not default)

DECISION D1 — primary obfuscating transport. SURFACED, not silently resolved [SYNTHESIS §3 D1, 04_ARCH §0, 11_MST].

- **Camp A (recommended, CLD/04_ARCH):** *masque-h3* is the primary obfuscating transport. It is *true Mullvad parity* (Mullvad's QUIC mode IS WG-over-MASQUE) and a **single Rust implementation** shared client[?]edge (I4). WG stays the crypto core unchanged.
- **Camp B (plurality of the 10-LLM analyses):** **Hysteria2 + Salamander** (QUIC + obfs, turnkey, BBR/Brutal congestion) as the primary obfuscating transport with WG as fallback — *ships faster* because Hysteria2 is a mature, turnkey QUIC-obfs stack.

Recommendation: adopt **Camp A (masque-h3) as the primary** for MVP, because (1) it preserves the "WG crypto core, pluggable transport" invariant (a Hysteria2-primary design makes Hysteria2 *the* protocol, not a carrier under WG, weakening I1/I4); (2) it is the only path that is literal Mullvad parity; (3) it keeps one Rust transport crate. **Provide hysteria2 as an additional Transport impl behind a Cargo feature** so the turnkey QUIC-obfs path is available as a ladder rung and as a Phase-0 unblock if *masque-h3-in-Rust* proves painful (the documented G2 fallback: "prototype MASQUE first in Go to unblock G2, then port" [04_P0 §13]). Asymmetric-per-leg (decision D6, §11) can additionally place Hysteria2 on the user[?]gateway leg only [11_MST].

```
// helix-transport/src/hysteria2.rs - feature = "hysteria2"
#[cfg(feature = "hysteria2")]
pub struct Hysteria2Transport { conn: quinn::Connection, /*
    salamander obfs state */ health: HealthCell }
// Carries WG datagrams over Hysteria2's QUIC datagram channel +
    Salamander packet obfuscation.
// Reuses the SAME quinn dependency as masque-h3; the obfs
    differs, the WG-datagram contract is identical (I2).
```

4. helix-wg — the boringtun wrapper (the WG crypto core handle)

boringtun is the right choice: pure-Rust userspace WireGuard, no kernel module, runs identically on Linux/Android/iOS — exactly the iOS path (no kernel WG on iOS regardless) [04_P0 §4.4]. On Linux the kernel WG fast path is used where

available, boringtun userspace as fallback (decision: ship both, default to kernel) [04_ARCH §13]. The wrapper owns WG's Tunn state machine, pumps its timer, and routes its four output verdicts.

```
// helix-wg/src/lib.rs (sketch)
pub struct WgPeer { tunn: boringtun::noise::Tunn, /* keys,
    endpoint, allowed_ips */ }

pub enum WgVerdict<'a> {
    WriteToTransport(&'a [u8]), // encrypted datagram → hand to
    the Transport (§3)
    WriteToTun(&'a [u8]), // decrypted IP packet → hand to
    the TUN
    Nothing,
    Err(WgError),
}

impl WgPeer {
    /// Inbound: bytes arrived from the transport → maybe
    plaintext for the TUN,
    /// or a handshake reply to send back out the transport.
    pub fn handle_transport_in(&mut self, datagram: &[u8],
        scratch: &mut [u8]) -> WgVerdict { /* ... */ }
    /// Outbound: a plaintext IP packet from the TUN → encrypted
    datagram for transport.
    pub fn handle_tun_out(&mut self, ip_pkt: &[u8], scratch: &mut
        [u8]) -> WgVerdict { /* ... */ }
    /// Call ~every 100–250 ms: emits keepalives / rekeys.
    pub fn tick(&mut self, scratch: &mut [u8]) -> WgVerdict
        { /* ... */ }
}
```

Crypto parameters are fixed by WireGuard and never altered [04_ARCH §2.2, SYNTHESIS §2]: Noise IK handshake, Curve25519 ECDH, ChaCha20-Poly1305 AEAD, BLAKE2s. Device private keys are generated on-device and **never leave the device** (doc 02 enrollment) [04_ARCH §7, SYNTHESIS §7]. Post-quantum is an additive **PSK** layer (ML-KEM/FIPS-203, hybrid-never-PQ-only) in Phase 2 — it slots into Tunn's preshared-key field with no transport change [04_P2, SYNTHESIS §4]; specified in the security doc, noted here only as a non-breaking WG-layer seam.

5. Orchestrator, status stream, and auto transport selection

5.1 The three loops + the connector posture

The orchestrator (crates/helix-core) runs three loops [04_P0 §4.5]:
tun → wg → (daita) → transport, transport → wg → tun, and timer tick.
The **connector** is the *same core* with a different routing posture: instead of capturing the device's default route it (a) advertises its served CIDRs (e.g. 10.10.0.0/24) in its map, and (b) NATs/forwards decapsulated packets into the LAN interface and back. One binary, a `--mode={client|connector}` flag [04_P0 §4.5].

client: [app traffic] → TUN → WG encrypt → [DAITA] → Transport
→ gateway
connector: gateway → Transport → WG decrypt → forward into
served CIDR (and reverse)

5.2 The status enum (the real-time UX seam)

The orchestrator exposes an event stream via **tokio::sync::broadcast** — the same enum the FFI surfaces to Flutter (doc 05) and what proves the real-time status UX [04_P0 §4.5, §9].

```
// helix-core/src/status.rs (mirrored across the FFI boundary,
// doc 05)
#[derive(Clone, Debug)]
pub enum TunnelStatus {
    Connecting,
    Handshaking,
    Connected { transport: String, rtt_ms: u32 }, // e.g.
        "masque-h3", 23
    Reconnecting,
    Down { reason: String },
}

pub struct Orchestrator {
    status_tx: tokio::sync::broadcast::Sender<TunnelStatus>, //
        fan-out to FFI + metrics + ladder
    // ... wg peers, active transport, daita stage, reconciler
    // handle ...
}

impl Orchestrator {
    pub fn subscribe(&self) ->
        tokio::sync::broadcast::Receiver<TunnelStatus> {
        self.status_tx.subscribe() }
}
```

5.3 Auto transport selection — the escalation ladder

Selection is **automatic with manual override** — Mullvad's exact UX [04_ARCH §3.2]. The client walks `TransportPolicy.order` on repeated handshake failure; each rung has a failure budget (N handshakes / T seconds) before escalating [04_P2 §1.4].

```
// helix-core/src/ladder.rs (sketch)
pub struct TransportPolicy {
    pub order: Vec<TransportKind>, // pushed by coordinator;
        e.g. regional prior (§5.4)
    pub pin: Option<TransportKind>, // user manual override –
        skip the ladder entirely
    pub budget: FailureBudget, // { max_handshakes: u8,
        window: Duration } per rung
}

// Default order (unrestricted network):
```

```
// [ plain-udp ]
// Restricted prior (e.g. censored region, pushed by coordinator):
// [ plain-udp, lwo, masque-h3, shadowsocks, udp-over-tcp ]
```

Selection algorithm [04_ARCH §3.2, 04_P2 §1.4]:

1. Start at pin if set (manual override), else order [0].
2. dial() the rung; on Connected within budget → done; emit Connected{transport, rtt}.
3. On DialTimeout/HandshakeFailed/EndpointBlocked exceeding the rung's failure budget → escalate to the next rung; emit Reconnecting.
4. On success, **remember the working transport per-network** (SSID / gateway fingerprint) so reconnects on the same hostile network skip straight to what worked — the difference between "VPN that sometimes connects" and "VPN that just works" [04_P2 §1.4].
5. The coordinator may push a **regional prior** (e.g. clients from CN-resolved IPs start at shadowsocks) via TransportPolicy.order, so users in censored regions do not pay the escalation latency every time [04_P2 §1.4].
6. Telemetry records **only** "transport X succeeded after N escalations in region R" (aggregate, no per-user data, I5) → feeds the Censorship-Evasion Success dashboard [04_ARCH §9, 04_P2 §1.4].

5.4 Sequence diagram — the transport escalation ladder

```
sequenceDiagram
    autonumber
    participant App as Client app (Flutter)
    participant Orch as Orchestrator (helix-core)
    participant Lad as Ladder (TransportPolicy)
    participant Tx as Transport (dial)
    participant Edge as Gateway edge (:443)

    App->>Orch: start(transport = auto)
    Orch->>Lad: select(order = [plain-udp, lwo, masque-h3, shadowsocks, udp-over-tcp])
    Orch-->>App: status = Connecting

    Note over Lad,Edge: Rung 1 – plain-udp (default fast path)
    Lad->>Tx: dial(PlainUdp)
    Tx->>Edge: WG handshake over UDP :51820
    Edge-->Tx: dropped (DPI blocks WG/UDP) → HandshakeFailed
    Tx-->>Lad: Err(HandshakeFailed) ; budget exceeded

    Note over Lad,Edge: Rung 2 – lwo (cheap header obfs)
    Orch-->>App: status = Reconnecting
    Lad->>Tx: dial(Lwo)
    Tx->>Edge: WG handshake over mangled UDP
    Edge-->Tx: dropped (signature still caught) → HandshakeFailed
    Tx-->>Lad: Err(HandshakeFailed) ; budget exceeded

    Note over Lad,Edge: Rung 3 – masque-h3 (CONNECT-UDP / HTTP-3 on :443)
    Lad->>Tx: dial(MasqueH3)
```

```
Tx-->Edge: QUIC/H3 CONNECT-UDP (RFC 9298) on :443/udp
Edge-->Tx: 200 (flow established) ; looks like HTTP/3
Tx-->Edge: WG datagrams as HTTP Datagrams (RFC 9297/9221)
Edge-->Tx: WG handshake reply
Tx-->Lad: Ok(Connected, rtt = 23 ms)
Lad-->Lad: remember per-network: this SSID → start at masque-
h3 next time
Orch-->App: status = Connected { transport: "masque-h3",
rtt_ms: 23 }
```

Note over Orch,Edge: if masque-h3 also fails → escalate to shadowsocks → udp-over-tcp

6. Overlay addressing & multi-network routing

This is the part the original ops runbook never addressed [04_ARCH §3.4]. A single user reaches **N joined private networks**, which may have **colliding RFC1918 ranges** (two connectors each serving 192.168.1.0/24) — that collision **MUST** be solved at v1 [SYNTHESIS §3 D4].

6.1 Overlay addressing — ULA /48 + 4via6 (decision D4, recommended)

DECISION D4 — IP-subnet collision across N joined networks. SURFACED [SYNTHESIS §3 D4, 04_ARCH §3.4].

- **Camp A (recommended, 04_ARCH): IPv6 ULA /48 per tenant** (fd7a:helix:<tenant>::/48)
 - Tailscale-style **4via6** mapping of advertised IPv4 LANs, so overlapping 192.168.1.0/24s across different connectors never collide.
- **Camp B (GMI/KMI): CGNAT 100.64.0.0/10, 1:1 per network** — assign each joined network a distinct slice of the CGNAT space and 1:1-NAT into it.

Recommendation: adopt **Camp A (ULA /48 + 4via6)**. It scales to far more networks than a hand-partitioned CGNAT space, is the proven Tailscale model, and is IPv6-native (the future default). Surface 4via6 as the engine but **hide it behind Console UX** — 4via6 mapping is powerful but confusing to end users (a named open risk) [04_ARCH §13]. CGNAT 1:1 remains a documented fallback for IPv4-only edge environments.

- Every node (client, connector, advertised host) gets a **stable overlay address** in the tenant's ULA /48 [04_ARCH §3.4].
- An advertised IPv4 LAN 10.10.0.0/24 behind connector A is mapped to a deterministic IPv6 prefix via 4via6: fd7a:helix:<tenant>:<siteID>:<4via6-encoded-v4>. The siteID disambiguates two connectors that both serve 192.168.1.0/24: each gets a distinct overlay prefix, and the gateway NATs into the correct connector [04_ARCH §3.4 overlapping-CIDR handling].

6.2 Prefix advertisement & routing map

Connectors advertise their served CIDRs to the control plane; the control plane compiles a **routing map** (which connector is next hop for which overlay prefix) and pushes it to the gateway edge and to authorized clients via WatchNetworkMap (doc 03) [04_ARCH §3.4]. The data plane consumes the map into helix-route:

```
// helix-route/src/map.rs (the desired-state the reconciler
// converges to; doc 03 streams it)
pub struct RouteMap {
    pub self_overlay: IpAddr,           // this node's
    // overlay IP
    pub peers: Vec<PeerRoute>,         // already policy-
    // filtered (need-to-know, I6)
    pub dns: Vec<IpAddr>,
}
pub struct PeerRoute {
    pub wg_pubkey: [u8; 32],
    pub endpoint_candidates: Vec<SocketAddr>, // for §8 NAT
    // traversal
    pub allowed_ips:
    Vec<IpNet>, // overlay prefixes this peer is
    // the next hop for
    pub via_connector: Option<SiteId>, // 4via6 site
    // disambiguation
}
```

Phase-0 fakes the stream with a static map.json of the *exact shape* Phase 1's WatchNetworkMap streams, so the reconciler is real even though the source is static [04_P0 §10].

6.3 Reconciliation (push, don't poll)

Agents are **declarative reconcilers**: they diff desired-vs-actual RouteMap and converge — bring peers up/down, switch transport, update routes — **without restarting** [04_ARCH §4.4, 04_P0 §10]. Convergence target: a route/policy change reflected on all affected edges in < **1 second** [04_ARCH §4.4]. The reconciler never polls; it reacts to map deltas pushed on the open stream (file-watch stands in during Phase 0).

7. Policy / ACL → AllowedIPs + nftables/eBPF verdict map

Default-**deny** (I6). A declarative allow-list (Tailscale-ACL-flavored, group:contractors → net:warehouse-cameras:554/tcp) is compiled by the control plane and expressed at the data plane as **per-peer AllowedIPs plus an nftables/eBPF verdict map** on the edge [04_ARCH §3.4, §7, SYNTHESIS §4].

Two enforcement layers — **both** required (defense in depth):

1. **WireGuard AllowedIPs** (cryptographic routing): each peer's AllowedIPs is the set of overlay prefixes that peer is allowed to send/receive — WG itself drops a packet whose source does not match the peer's allowed set. This is the coarse, crypto-enforced layer.
2. **Edge verdict map** (port/proto-granular): an **nftables verdict map** (Phase 1) or **eBPF** program (Phase 2, scale) on the gateway evaluates the compiled policy rule (src_group → dst_selector : ports : action) per flow. Default-deny; fail-closed (if the compiled policy is absent/stale, drop) [04_ARCH §7, SYNTHESIS §4].

```
// helix-route/src/policy.rs (compiled verdict the edge installs)
pub struct CompiledPolicy { pub version: u64, pub rules:
    Vec<VerdictRule> }
pub struct VerdictRule {
    pub src: IpNet, pub dst: IpNet, pub proto: L4Proto, pub
    ports: PortRange, pub action: Verdict,
}
pub enum Verdict { Allow, Drop } // default Drop (I6)
pub enum L4Proto { Any, Tcp, Udp, Icmp }

# edge-installed nftables verdict map (Phase 1 representation;
eBPF equivalent in Phase 2)
table inet helix {
    map policy_v {
        type ipv6_addr . ipv6_addr . inet_proto . inet_service :
verdict
        elements = {
            fd7a:helix:1::a . fd7a:helix:1:warehouse::20 . tcp .
554 : accept
        }
    }
    chain forward {
        type filter hook forward priority 0; policy drop; #
DEFAULT-DENY (I6)
        ip6 saddr . ip6 daddr . meta l4proto . th dport vmap
@policy_v
    }
}
```

Properties [04_ARCH §3.4, §7]: **split-horizon** — connectors cannot reach each other unless policy says so; clients cannot reach networks they are not granted; microsegmentation is the default. Revocation: device.revoked (doc 02) → the verdict map drops the peer in **< 1 second** with no restart [04_ARCH §4.4, SYNTHESIS §7]. **Peers are delivered to each agent already policy-filtered** (need-to-know) so the client never even learns of networks it cannot reach (I6) [SYNTHESIS §7].

8. Direct peer-to-peer & NAT traversal (Phase 2 — additive)

Phase 1 relays **all** peer traffic through the gateway (simple, matches the proven slice). Phase 2 makes the gateway a **coordinator + relay-of-last-resort**, with traffic going **directly** between client and connector wherever NAT allows — lower latency, far less gateway bandwidth [04_P2 §3]. This is purely additive: it fills `PeerRoute.endpoint_candidates` (the field §6.2 reserves) and changes which endpoint WG latches onto — no transport or trait change.

1. **Endpoint discovery (STUN-like)** [04_P2 §3.1]: each node learns local candidates (all interface addresses) + server-reflexive candidates (probe the gateway's STUN-like endpoint, which echoes the observed `src ip:port`, revealing the NAT mapping). Candidates are reported to the coordinator and distributed to authorized peers as `endpoint_candidates`.
 2. **Hole punching** [04_P2 §3.2]: both peers simultaneously send WG handshake probes to all of the peer's candidates; for full-cone / restricted-cone / port-restricted NATs a path opens, and **WireGuard's roaming latches onto whatever source address a valid handshake arrives from** — no extra machinery. Symmetric-NAT-on-both-ends → hole punching fails → fall back to relay.
 3. **Signaling = the existing stream** [04_P2 §3.3]: *WatchNetworkMap* is the signaling channel (the coordinator pushes each peer the other's candidate list as a delta) — no separate signal service (HelixVPN folds NetBird-style signaling into the coordinator).
 4. **Relay fallback (DERP-style)** [SYNTHESIS §4]: when both ends are behind symmetric NAT, traffic relays through a *helix-relay* (gateway) — relay-of-last-resort, not the default.
-

9. DAITA — traffic-shaping stage (Phase 2, data-plane placement only)

DAITA defeats size/timing/frequency fingerprinting even of fully-encrypted flows. **Do not roll your own** — adopt **maybenot** (the framework behind Mullvad's DAITA): a state-machine engine for padding + cover-traffic "machines" [04_P2 §2, SYNTHESIS §4]. The data-plane spec owns only its **placement** and the per-packet hook; the privacy design (machines, off-by-default policy, website-fingerprinting validation harness) lives in the security doc.

Placement (L2.5): a stage **above WireGuard, below the transport** — it operates on the stream of WG datagrams (padding/injecting cover) before they hit the obfuscating transport, orthogonal to *which* transport runs [04_P2 §2.2].

TUN → WG encrypt → [DAITA: pad + inject cover] → Transport → wire

```
// helix-daita/src/lib.rs (sketch)
pub struct Daita { framework: maybenot::Framework /* machines are
    CONFIG from NetworkMap, not code */ }
impl Daita {
    /// Per outgoing WG datagram: may pad it and/or schedule cover
    packets.
```



```

pub fn on_packet(&mut self, dg: &mut Bytes, now: Instant) ->
    Vec<ScheduledAction> { /* ... */ }
/// Timer-driven: emit scheduled cover/padding packets.
pub fn tick(&mut self, now: Instant) -> Vec<Bytes> { /* ... */ }
}

```

Machines are distributed as **data** via a **daita** field on the NetworkMap, so defenses tune server-side without a client rebuild [04_P2 §2.3]. Off by default, opt-in ("maximum privacy" with an honest bandwidth/latency cost note) [04_P2 §2.3].

10. MTU & overhead budget

The transport changes how many bytes of WG payload fit per L1 packet; each impl reports its `effective_mtu()` and the orchestrator sets the inner WG MTU accordingly [04_P0 §4.2, §5.2].

Transport	<code>effective_mtu()</code>	Rationale	Source
plain-udp	1420	standard WG-over-IPv4 (1500 – 20 IP – 8 UDP – 32 WG hdr/tag ≈ 1420)	[04_P0 §4.3]
masque-h3	1280 (measure & tune)	QUIC + HTTP-Datagram + UDP-proxy overhead eats headroom; 1280 is IPv6 min-MTU floor	[04_P0 §5.2]
connect-ip	~1300	IP-over-H3, no inner-WG header but QUIC overhead	derived
shadowsocks	~1380	TCP MSS – 2-byte length prefix – AEAD tag	derived [04_P2 §1.1]
udp-over-tcp	~1380	TCP MSS – length prefix	derived [04_P2 §1.2]
lwo	~1400	plain-udp minus padding bytes	derived [04_P2 §1.3]

Rules: (1) the inner WG MTU = `min(active transport.effective_mtu(), path-MTU-discovered)`; (2) when DAITA pads, padding is added *after* WG encrypt and counts against the transport budget, not the inner MTU; (3) Oversize is a hard error (`TransportError::Oversize`) — the orchestrator must fragment at L3 (rare) or lower the inner MTU rather than silently truncate. The MASQUE MTU penalty is **quantified in Phase 0** (record MTU/throughput/CPU vs plain-udp) so the ladder's cost model is real [04_P0 §5.3].

11. Multi-hop (nested WireGuard) + edge-language & topology decisions

11.1 Multi-hop — nested WG with per-hop keys

Generalize the original "chain two VPSes" [04_ARCH §3.5]. A client may route Client → Gateway-Entry → Gateway-Exit → {internet | connector}. **Entry sees the client but not the destination; exit sees the destination but not the client** — the Mullvad multi-hop property. Implemented as **nested WireGuard with per-hop keys**: the client holds a WG session to Entry *and* a WG session to Exit; the Exit session's datagrams are themselves the payload of the Entry session. Orchestrated by the control plane and pushed as a multi-hop RouteMap (PeerRoute chain) [04_ARCH §3.5]. The transport layer is unchanged — each hop is an ordinary Transport carrying that hop's WG datagrams (the outer hop may be `masque-h3`, the inner `plain-udp`).

```
// helix-core: a hop chain is just an ordered Vec of WgPeer +  
//               Transport pairs  
pub struct HopChain { pub hops: Vec<Hop> }           // [Entry,  
//               Exit] for 2-hop  
pub struct Hop { pub wg: WgPeer, pub transport: Box<dyn  
//               Transport> }  
// outbound: encrypt for Exit (innermost) → encrypt for Entry  
//            (outermost) → Entry transport.send()
```

11.2 Decision D5 — gateway edge language (MASQUE termination)

DECISION D5 — gateway edge language. SETTLED BY PHASE-0 G4 BENCHMARK [SYNTHESIS §3 D5, 04_P0 §7].

- **Camp A (recommended, 04_P0): Rust** (quinn + h3 + hand-rolled CONNECT-UDP) — the edge shares helix-transport **byte-for-byte** with clients (I4); no GC on the hot path.
- **Camp B: Go** (quic-go + masque-go, turnkey CONNECT-UDP/IP) — matches the Go/Gin control plane, more mature MASQUE tooling; cost = dual MASQUE implementations to keep in sync.

Recommendation / decision rule [04_P0 §7.3]: choose **Rust** if it gets within ~10–15% on throughput/CPU-per-Gbps and the hand-rolled CONNECT-UDP is not a quagmire (the single-implementation win). Choose **Go** if MASQUE-in-Rust proves painful or Go wins decisively on cost-to-serve, and accept dual implementations mitigated by **shared test vectors + a conformance suite across both edges**. The call is recorded in the Phase-0 decision log with benchmark numbers (throughput @ 1/10/100 clients, CPU/Gbps, p99 latency incl. GC tail, handshake churn/sec, memory under churn) [04_P0 §7.2, §8]. **This spec assumes Rust** and notes the Go path as the conformance-suite-mitigated fallback.

11.3 Decision D6 — transport topology (per-leg)

DECISION D6 — transport topology. SURFACED [SYNTHESIS §3 D6, 11_MST].

- **Default (04_ARCH):** a single protocol family end-to-end (WG everywhere, the obfuscating transport chosen per the ladder).
- **Camp B (11_MST, distinctive): asymmetric per-leg** — Hysteria2/QUIC on the *user*→*gateway* leg (where obfuscation matters most), plain WireGuard on the *gateway*→*networks* leg (where the connector dials out of a cooperative network).

Recommendation: the Transport trait already makes per-leg topology free — each leg is an independent Transport selection. Adopt **best-fit-per-leg as a policy, not a new mechanism**: the coordinator pushes a `TransportPolicy.order` per leg, so the *user*→*gateway* leg can start at `masque-h3/hysteria2` while the *gateway*→*connector* leg stays `plain-udp` on cooperative networks. No code change beyond per-leg `TransportPolicy` — surface it as the default Phase-2 behavior.

12. File-by-file build checklist (data plane)

Phase-tagged; each item maps to a workable item in the §11.4.93/.95 SQLite tracker [SYNTHESIS §9].

Crate / file	Owns	Phase	Phase-0 gate
helix-transport/src/lib.rs	Transport trait, <code>TransportConfig</code> , <code>dial()</code> , registry	0	S0
helix-transport/src/plain_udp.rs	baseline UDP carrier	0	G1 / S1
helix-transport/src/masque.rs	CONNECT-UDP / HTTP-3 / QUIC datagrams	0	G2 / S3
helix-transport/src/error.rs	<code>TransportError</code> taxonomy	0	S0
helix-wg/src/lib.rs	<code>boringtun</code> Tunn wrapper, <code>WgVerdict</code>	0	G1 / S1
helix-tun/src/lib.rs	Linux TUN; fd-injection for shims	0	S1
helix-core/src/orchestrator.rs	three loops, client/connector posture	0	S2
helix-core/src/status.rs	<code>TunnelStatus</code> broadcast enum	0	G5 / S5
helix-core/src/ladder.rs	<code>TransportPolicy</code> , auto-selection	1	—
helix-route/src/map.rs	<code>RouteMap</code> reconciler (push, no restart)	0→1	G6 / S8

Crate / file	Owns	Phase	Phase-0 gate
helix-route/src/policy.rs	ACL→AllowedIPs+verdict-map compiler	1	—
helix-transport/src/shadowsocks.rs	SS-wrap carrier	2	—
helix-transport/src/udp_over_tcp.rs	UoT last-resort	2	—
helix-transport/src/lwo.rs	lightweight obfs (basic P1, hardened P2)	1→2	—
helix-transport/src/connect_ip.rs	RFC 9484 IP-over-H3 (feature-gated)	2	—
helix-transport/src/hysteria2.rs	Hysteria2+Salamander carrier (feature-gated, D1)	1→2	—
helix-daita/src/lib.rs	maybenot shaping stage	2	—
helix-core/src/hop_chain.rs	nested-WG multi-hop	2	—
bin/helix-edge.rs	gateway MASQUE termination (D5)	0	G4 / S4

12.1 Anti-bluff acceptance evidence (constitution §11.4.5/.69/.107, SYNTHESIS §9)

Every data-plane claim ships captured evidence, not config-only PASS:

- **G1 plain-UDP slice** — iperf3 through-tunnel $\geq 80\%$ bare link + ping to LAN host succeeds; CSV from bench.sh [04_P0 §0, §8].
- **G2 MASQUE through DPI block** — slice works with plain WG **blocked** (nft rule), tshark capture classifies the flow as HTTP/3 with **no WG signature**, goodput @ 5% netem loss beats the UoT strawman [04_P0 §5.3, §8].
- **G6 reconcile** — edit the static map → peer reachable, **no restart** of unrelated state [04_P0 §10].
- Per-transport: effective_mtu() measured (not assumed); ladder escalation captured as an ordered TunnelStatus event trace; verdict-map default-deny proven by a denied-flow capture.

13. What is explicitly out of scope for this document

Control-plane services, the WatchNetworkMap protobuf/Connect contract, Postgres data model & RLS, enrollment/PKI, event bus (doc 02/03); client UI, design system, FFI Dart bindings (doc 05); platform tunnel shims beyond the TUN-fd handoff contract — iOS NEPacketTunnelProvider memory ceiling (G3, the make-or-break gate), Android VpnService/JNI, Windows wireguard-nt, HarmonyOS Network Kit, Aurora Qt/tun (doc 06); deployment quadlets, observability, HA (doc 07/08). The post-quantum PSK, DAITA privacy policy, and kill-switch/DNS-leak state machine are *referenced* here for their data-plane seams and *specified* in the security doc.

End of data-plane specification (document 01). Pair with doc 03 (WatchNetworkMap wire contract — the live source of the static map.json seam at §6.2) and doc 06 (platform tunnel shims — the TUN-fd handoff at §5.1). Surviving interfaces from Phase 0 (the Transport trait §3.1, helix-wg §4, the TunnelStatus enum §5.2, the RouteMap §6.2) are frozen contracts; their implementations may evolve [04_P0 §14].